

Assignment: Unsupervised learning - Clustering

DSA 8401: Applied Machine Learning

Master in Data Science and Analytics



Strathmore University

@iLabAfrica Centre

1 Goals

This week's activity focuses on practising one of the most popular unsupervised learning methods: Clustering. It can be useful when we begin the analysis of a data set and we want to start getting to know our data, exploratory analysis. Other important application of clustering is for stratify groups of people, for instance patients. So the the goal of this assignment is to stratify patients.

2 Assignment Description

In this activity we will work with the dataset called "Pima Indians Diabetes", which you can find at <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>.

It is requested:

1. We will download and preprocess the diabetes dataset following the suggestions given week 2.
2. Create a dataframe called diabetesDF based on the dataset downloaded. Analyze it and observe the attributes.
3. From here you have to explore the data and extract all the knowledge you can. Follow these steps:
 - (a) Considering all the attributes of the dataset, except for the 'Outcome'. Create a normalized version of it (saved in a numpy array called dataClust).
 - (b) Using the k-means Clustering algorithm provided by the sklearn Python library, we generate a graphical representation of the SSE (sum of squared errors) vs. number of clusters. If we wanted to use the elbow rule seen in theory, how many clusters would we use? Is it a practical value?
 - (c) Now we apply the k-means algorithm to generate 3 clusters. Once created, analyze the centers of the clusters obtained. We now show the centers of each cluster and the number of elements in each of them. What attributes determine the most notable differences between clusters?
 - (d) We now add to the original data frame (DiabetesDF) a new 'Cluster' **attribute** indicating which Cluster each element belongs to.

- (e) We analyzed the centers of each cluster again but using DiabetesDF (non-normalized values) to discern the influence of each attribute on clusterization. Try to describe what characterizes each cluster. Remember that the attributes that best discriminate are those that make one centroid more separated from another.
- (f) Finally, we give meaningful names to the clusters that reflect the types of patients. That is, if you had to describe the patients in each cluster with 2-3 keywords, how would you do it?
- (g) Use Gaussian Mixtures to generate the clusters, comparing the result with the previous case. Discuss what you observed and reason which option you think is most appropriate.

3 Hand in

The students will upload in the eLearning platform the Jupyter notebook. The notebook will contain the python code, the plots and especially personal responses to the previous questions. If there is no personal comment the mark will be F. Marks C,B and A will depend on the variety and quality of the personal comments.